

Irrigation Modernization - Phase 2

Finca Miles - parcel N-2, flood to precision drip

Parcel	N-2 (Norte Bajo), 5.5 ha	Project total	EUR 47,800
Conversion	Flood to precision drip	EU grant (40%)	EUR 19,120
Contractor	Riegos del Sur	Net to estate	EUR 28,680
Quote ref	RDS-2026-339	Scheduled	2026-10-05

1. Background

The estate is modernizing irrigation parcel by parcel to cut water use and improve fruit quality and yield stability under an increasingly hot, dry Mediterranean climate. Phase 1 converted parcel N-1 (6 ha) to drip in November 2025 on a HydroSense smart controller, completing its first full season this year. Phase 2 converts the adjacent N-2 (5.5 ha), completing drip conversion of the north grove.

2. Objectives

- Replace flood irrigation on N-2 with pressure-compensating drip for uniform application.
- Integrate N-2 into the existing HydroSense controller and soil-moisture telemetry.
- Reduce applied water per hectare and enable deficit-irrigation strategies at fruit set.
- Improve yield stability and oil quality through controlled water stress.

3. Scope of work

Installation of drip lines with pressure-compensating emitters, submains and manifolds, a filtration and fertigation station, a pump and variable-speed-drive upgrade, six soil-moisture sensors with telemetry, and integration into the HydroSense controller; commissioning and as-built documentation on completion.

4. Bill of quantities (summary)

Item	Qty	Line EUR
Drip line 16mm PE + pressure-comp emitters	8,200 m	11,890
Submains and manifolds	lot	6,800
Filtration + fertigation station	lot	7,400
HydroSense controller integration	lot	3,200
Trenching and installation labour	lot	12,600
Pump + variable-speed drive upgrade	lot	4,100
Soil-moisture sensors (6) + telemetry	6 units	1,410
Commissioning and as-built	lot	400
TOTAL		47,800

Full priced bill of quantities is in workbook X06.

5. EU grant (FEADER rural development)

Field	Value
Programme	FEADER rural development (Andalusia)
Measure	M04.1 (on-farm investment)
Grant reference	FEADER-AND-2026-01188
Co-financing rate	40%
Project total	EUR 47,800
Grant amount	EUR 19,120
Net to estate	EUR 28,680

6. Schedule

Milestone	Date
Contractor confirmed (Riegos del Sur)	Jan 2026
Grant approval	2026
Installation start	2026-10-05
Commissioning	Oct-Nov 2026
First full drip season (N-2)	2027

Installation is timed for the dry dormant period after summer and before the 2027 growing season, avoiding the harvest window.

7. Expected benefits

- Lower applied-water volume per hectare versus flood irrigation.
- More uniform tree water status and better fruit quality.
- Completion of north-grove drip conversion (N-1 + N-2).
- Data-driven scheduling via soil-moisture telemetry and the HydroSense controller.

8. Water and climate context

The Subbética olive country faces hotter summers and less reliable rainfall. Flood irrigation is water-hungry and uneven; precision drip applies water where and when the tree needs it and enables controlled deficit irrigation, which can concentrate quality without stressing the orchard. Modernizing irrigation is the estate's single most important climate-adaptation investment.

9. Phase 1 results (N-1, reference)

Measure	Phase 1 (N-1)
Area	6 ha drip, Nov 2025
Controller	HydroSense (smart)
First full season	2025-26 complete
Outcome	uniform application; baseline for N-2

10. Operation and maintenance

- Seasonal flushing of drip lines and filter cleaning.
- Emitter checks and pressure verification each spring.
- Sensor calibration and telemetry review through the HydroSense controller.
- Spare sensor and seal stock held (see equipment workbook X12).

11. Approvals

Any commitment at or above the approval threshold proceeds only on the owner's explicit confirmation; the contractor (Riegos del Sur) was selected on price and local reputation after comparing two quotes.

Prepared by the office of Min Miles Park, Finca Miles. Costing: workbook X06.